

The story of the atom

Atoms are the building blocks of matter. These incredibly tiny units typically measure just one tenth of a billionth of a meter. The word “atom” comes from the Greek *atomos*, meaning indivisible. In the 19th century, atoms were considered to be the smallest units of matter, but advances in atomic science have revealed an inner structure made of even smaller particles. Every element has a unique type of atom. So far, 118 different elements have been discovered (see pp.188–189).

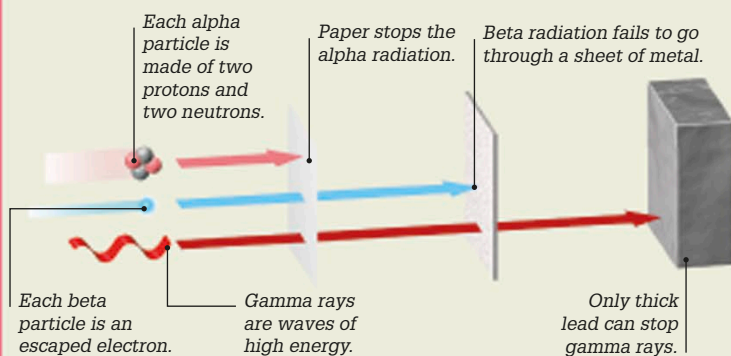
Neutrons do not have any charge.

Atomic structure

Atoms are made of particles called protons, neutrons, and electrons. Protons and neutrons form the central part of the atom—the nucleus. Protons carry a positive electrical charge, which attracts the negatively charged electrons and keeps them in orbit around the nucleus.

Radioactivity

An atom is radioactive if its nucleus is unstable and breaks apart and decays, emitting energy and particles known as radiation. Decay occurs at a fixed rate and the time it takes for half the mass of a radioactive substance to decay into other elements is called its half-life.



Types of radiation

The decay of a radioactive atom involves the emission of three main types of radiation: alpha, beta, and gamma. Alpha radiation (a stream of alpha particles) travels only a few centimeters in air and cannot get through a sheet of paper, while beta radiation can travel further but bounces off thin sheets of metal. Gamma radiation can penetrate many materials.

Protons are found in the nucleus, which is normally stable.

Key events

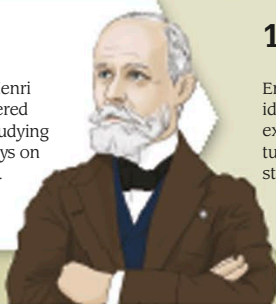
c 400 BCE

Ancient Greek philosopher Democritus described how matter is made up of small, indivisible particles, which have different forms and arrangements. He called these particles atoms.

1896

French scientist Henri Becquerel discovered radiation while studying the effects of X-rays on photographic film.

Henri Becquerel



1897

English physicist J. J. Thomson identified the electron while experimenting with cathode ray tubes. His discovery was the first step in the study of the atom.

1913

Danish scientist Niels Bohr proposed a model of the atom in which electrons occupy shells, or orbits, of differing energy around the nucleus.